Student Name: ____

FURTHER MATHEMATICS 2016

Unit 3
Key Topic Test 5 – Recursion and Financial Modelling: Depreciation

Recommended writing time*: 45 minutes
Total number of marks available: 25 marks

QUESTION BOOK

* The recommended writing time is a guide to the time students should take to complete this test. Teachers may wish to alter this time and can do so at their own discretion.

Conditions and restrictions

- Students are permitted to bring into the room for this test: pens, pencils, highlighters, erasers, sharpeners and rulers.
- Students are NOT permitted to bring into the room for this test: blank sheets of paper and/or white out liquid/tape.
- One approved graphics calculator or approved CAS calculator or CAS software and, if desired, one scientific calculator.

Materials supplied

- Question and answer book of 8 pages.

Instructions

- Print your name in the space provided on the top of the front page.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic communication devices into the room for this test.

SECTION A – Multiple-choice questions

Instructions for Section A
Choose the response that is **correct** for the question and circle the correct answer.
A correct answer scores 1, an incorrect answer scores 0.
Marks will not be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Question 1
The first three terms generated from the recurrence relation $V_{n+1} = 0.80 \times V_n + 100,\ V_0 = 200$ are:

A. $260,\ 308,\ 346.4$

B. $200,\ 260,\ 308$

C. $200,\ 300,\ 400$

D. $200,\ 160,\ 128$

E. $200,\ 380,\ 560$

The following information relates to Questions 2 and 3

A machine depreciates by 10% each year. Its initial value is $4800

Question 2
The recurrence relation that represents the value of the machine after *n* years is:

A. $V_{n+1} = 0.90 \times V_n,\ V_0 = 4800$

B. $V_{n+1} = 0.10 \times V_n,\ V_0 = 4800$

C. $V_{n+1} = 1.10 \times V_n,\ V_0 = 4800$

D. $V_{n+1} = 10 \times V_n,\ V_0 = 4800$

E. $V_{n+1} = V_n - 10,\ V_0 = 4800$

Question 3

The value of the machine after 4 years is:

A. $3499.20

B. $3888.00

C. $3149.28

D. $3499.20

E. $4440.00

The following information relates to Questions 4 and 5

Lee invests $6540 at 6.2% interest per annum compounded quarterly.

Question 4

The recurrence relation that will find the amount of interest he will earn in the n^{th} period of investment is given by $V_{n+1} = a \times V_n,\ V_0 = b$. The values of *a* and *b* respectively are:

A. $a = 0.062,\ \ b = 6540$

B. $a = 0.62,\ \ b = 6540$

C. $a = 1.55,\ \ b = 6540$

D. $a = 0.0155,\ \ b = 6540$

E. $a = 1.0155,\ \ b = 6540$

Question 5

The amount of his investment after 4 periods of investment is:

A. $6744.31

B. $6848.85

C. $7046.85

D. $8567.40

E. $6955.01

SECTION B- Short-answer questions

Instructions for Section B
You need not give numerical answers as decimals unless instructed to do so. Diagrams are not to scale unless specified otherwise.

Question 1

Leigh decides to buy new car for \$45000. Leigh knows that the acre will depreciate by 15% per annum on the reducing value of the car.

a. Find the value of the car

i. After one year

____

____

____

ii. After two years

____

____

b. Write a recurrence relation that represents the value of the car after *n* years.

____

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c. By how much has the value of the car reduced by in 4 years' time?

d. After how many years does the value of the car becomes less than quarter of the original value?

Leigh also decides to buy computer equipment for \$45000. The computer equipment depreciates at a flat rate of \$6000 per year.

e. Write down the value of the equipment in each of the first four years.

f. Write down the recurrence relation that calculates the value of the equipment after *n* years.

g. After how many years will the value of the equipment be first less than the value of the car?

$2 + 2 + 2 + 1 + 2 + 2 + 2 = 13$ marks

Question 2

Leigh invests \$20000 in a bank. The bank offers 5.4% per annum compounded monthly. Leigh also deposits \$200 at the end of each month.

a. What does Leigh's investment after 3 months?

b. Write down a recurrence relation for his investment.

c. Use the recurrence relation above to find the amount of investment after one year.

d. Find the amount of interest he receives after one year.

2 + 1 + 2 + 2 = 7 marks

END OF KEY TOPIC TEST